Tank Battle

James Smith - Game Design Document
Assignment #1
Due: 20 October 2026 (Tuesday)

Executive Summary

Tank Battle is a 2D, scrollable, top-down online tank war where four color teams raid each other's depots for enemy-colored Energy Cells, and the team with the most Cells in its depot after five rounds wins.

Up to 20 players share one large scrolling world. When you join, pick a tank color (Blue/Water, Red/Fire, White/Ice, or Gray/Rock); everyone with your color is your team, and every other color is an enemy. No lobby or tutorial to wait in: the moment a round starts, you are in the action. Two kinds of AI tanks share the map. Auxiliaries wear a team color and fight for that team. Wreckers are black, belong to no team, and hunt whoever carries the most Cells or who they come across in open terrain. Wreckers can also attack depots, but only two can fire on a depot; the rest sit and wait or attack other defenders. If a depot is destroyed, the team has to go and gather more Cells.

The hook is that every team owns an element. Blue tanks can only cross water, Red only fire, White only ice, and Gray only rocks; those same terrains hide tanks from Wreckers. A team can vanish into its own element, but Cells are only worth something if you carry them out into the open and back to your depot, where Wreckers look for the biggest haul. Greed and safety pull against each other every minute of play. All players, Wreckers, and Auxiliaries can cross grass (Green) and ground (dirt) (Brown).

- **Core loop:** raid an enemy depot, carry its Cells back to your own depot to store them, survive enemies and Wreckers, refuel, and go again, across 5 rounds of at most 7 minutes each (35 minutes maximum).
- **Win:** the team with the most stored Cells at the end of round 5, or the last team standing.
- **Loss:** a destroyed player is out until the next round; a team whose members are all destroyed in a round is removed from the game.

Game Mechanics

Each round, you move, aim, and fire at tanks, steal Cells from enemy depots, and refuel at stations, all while staying within your color's terrain.

What the player does

Action	What you see and do
Join	Pick Blue, Red, White, or Gray. Your tank appears in the live world, and the round is already running.
Move and aim	Steer (WASD) across the scrolling map; the view follows your tank. Open ground is passable for everyone, but each element is a wall to every color except its own.
Fire	Shoot enemy tanks, Auxiliaries, and Wreckers (Space Bar). Every target has 5 energy bars, and 3 hits drain one bar.
Raid	Drive into an enemy depot and pick up Cells in that depot's color. Your carried-Cell count shows on screen and rises as you steal, but the Cells do not score until you store them.
Store	Carry the stolen Cells back to your team's depot to store them. Only stored Cells count toward your team's total, and Cells you are still carrying can be lost if you are destroyed.
Defend	Guard your depot, where your team stores its Cells, because the other three teams are coming for your color's Cells.

Action	What you see and do
Refuel	Drive onto a refueling station to refill all 5 energy bars. Only online players can refuel.
Hide	Slip into your own element to break Wrecker line of sight and wait out a hunt.

One world per color

Team	Can cross	Cannot cross	Hiding terrain (blocks Wrecker sight)
Blue	Water	Fire, ice, rocks	Water
Red	Fire	Water, ice, rocks	Fire
White	Ice	Water, fire, rocks	Snow piles
Gray	Rocks	Water, fire, ice	Rock mounds

Because enemy colors cannot follow you into your element, the map is effectively four overlapping worlds. A raid usually means leaving your safe terrain, grabbing Cells on open ground, and getting back before something catches you.

Survival and elimination

A tank is destroyed after 15 hits without a refuel (5 bars at 3 hits each). A destroyed player is removed from the round and watches until the next one starts. If a team loses all its members in the same round, that team is removed from the game. If only one team is left, the game ends and that team wins.

The round loop

1. The round starts at 7:00 on the clock, and all online players spawn and act immediately.
2. Teams raid enemy depots and carry the Cells back to their own depot to store them. A new Wrecker appears every 15 seconds, so up to 28 can join the hunt by the end of a round.
3. Players trade fire, refuel, hide, and push again as the map gets more dangerous.
4. The round ends when the timer runs out or only one team remains, and stored Cells are tallied.
5. Destroyed players re-enter for the next round; eliminated teams do not.
6. After round 5, the team with the most stored Cells wins, so a match lasts at most 35 minutes.

Design Decisions

- Only stored Cells are tallied, cumulatively across all five rounds, and a destroyed carrier drops its unstored Cells where it falls.
- Auxiliaries use the same 5-bar, 3-hit durability as Wreckers and players.
- Team elimination counts human players only, not Auxiliaries.
- Players can only refuel while online, so disconnected players return at their last energy level.

AI Architecture

Tank Battle has two kinds of AI tanks: Wreckers punish whoever is winning, and Auxiliaries keep every team in the fight even when few players are online.

Wreckers (black, no team)

A Wrecker looks like a black tank with 5 energy bars. It cannot refuel; a destroyed Wrecker returns at a different spot on the map, and a fresh one arrives every 15 seconds. Its single mission is to destroy everyone else.

- **It hunts the Cell leader.** Whoever is carrying the most unstored Cells becomes the target, so a big haul makes you the most hunted tank on the map. This forces a choice between hiding and running for your depot, and gives trailing teams room to catch up.
- **It attacks anyone it sees.** A Wrecker chasing the leader will still fire on any other team it spots on the way, so open ground is dangerous for everyone.
- **It cannot see through water, fire, Snow piles, or Rock mounds.** A tank inside its own element is invisible to it. Players learn to hide while a Wrecker passes, then dash out the other side.
- **It wears down.** With no refueling, a Wrecker dies after 15 hits, so a team that focuses fire can clear it, but the next one is only seconds behind.

Auxiliaries (team color, fight for their team)

An Auxiliary looks like a player tank in your team's color so that you can tell friend and foe at a glance.

- **It defends and fights.** It shoots enemy tanks it sees and holds ground near its own depot, so an enemy raider meets resistance even when no human teammate is nearby.
- **It obeys the same terrain rules as its color.** A Blue Auxiliary can cross water but is stopped by fire, ice, and rock, which means enemies can predict and outmaneuver it.
- **It fills out small teams.** When fewer than 20 people are online, Auxiliaries stop a team from being an easy target and keep all four colors present on the map.
- **It can be shot.** Enemy players clear Auxiliaries to open a depot, giving raids an early objective before the Cells.

How each AI behaves, in plain terms

AI	What you notice	Rule behind it
Wrecker	Drifts across the map, then suddenly turns toward one tank and closes in	Roam until a Cell leader is known, hunt that tank, fire on anyone else in sight. Can attack depots.
Wrecker	Loses you when you duck behind a terrain barrier	Line of sight is cut by water, fire, snow, and rock
Auxiliary	Teammates' color tanks guarding your depot and engaging raiders	Defend depot, attack visible enemies, stay inside its color's terrain

Both AI types use fixed rules that run on the game server, not live calls to a language model. Section 4 explains why.

Agents that build the game

Sixteen single-purpose agents, each run in Claude Code, design, develop, and test Tank Battle. Each does exactly one job.

Agent	Its one job
Lead Agent	Hands each task to the right agent and checks that their pieces fit together.
Game Logic Agent	Writes the rules for energy bars, 3-hit damage, carrying and storing Cells, elimination, and round scoring.
World Map Agent	Builds the scrolling map with depots, refueling stations, and the water, fire, ice, and rock zones.
Game Control Agent	Turns player input into tank movement, aiming, and firing.
Wrecker AI Agent	Writes how black tanks roam, hunt the Cell leader, and lose sight of tanks hiding in terrain.
Auxiliaries AI Agent	Writes how team-color AI tanks defend their depot and attack visible enemies.
Graphics Agent	Draws the tanks in five colors, Cells, depots, stations, and terrain tiles.

Agent	Its one job
UI Agent	Designs the energy bars, Cell counter, round timer, and team color display.
Multiplayer Sync Agent	Keeps all 20 players seeing the same world and keeps the 7-minute round timer on the server.
Playtest Agent	Runs scripted matches and reports bugs, such as a round that runs past 7 minutes.
Collision and Terrain Agent	Decides what blocks what: each element as a wall to the other three colors, shells hitting tanks, and terrain cutting Wrecker line of sight.
Session and Join Agent	Handles picking a color, the 20-player cap, returning destroyed players at the next round, and refueling for online players only.
Audio Agent	Creates the sounds for firing, hits, destruction, refueling, and each new Wrecker arriving.
Deployment Agent	Sets up and keeps the persistent online game server running.
Effects Agent	Animates explosions, shell impacts, and the moving water, fire, snow, and rock terrain.
Balance Agent	Tunes the numbers that shape play, such as 3 hits per bar, the 15-second Wrecker spawn, and the 7-minute round.

Technical Strategy

Tank Battle is built using Claude Code by the sixteen single-purpose agents listed under AI Architecture, with a combined budget of about 1.9 million tokens, run in four waves so the project stays inside API rate limits.

Agent roles and token budget

Each agent runs as its own Claude Code session. Models are matched to difficulty: Opus 5.5 for the two jobs where mistakes cost the most (coordination and multiplayer sync), Sonnet 5.5 for gameplay code and AI, and Haiku 5.5 for simple, well-defined work. Budget equals tokens per task times the number of tasks, input plus output.

Agent	Model	Tokens/Task	Budget (tokens)	Cost
Lead Agent	Opus 5.5	20,000	120,000 (6 tasks)	$0.96
Game Logic Agent	Sonnet 5.5	25,000	150,000 (6 tasks)	$0.60
World Map Agent	Sonnet 5.5	22,500	90,000 (4 tasks)	$0.36
Game Control Agent	Haiku 5.5	17,500	70,000 (4 tasks)	$0.01
Wrecker AI Agent	Sonnet 5.5	26,000	130,000 (5 tasks)	$0.52
Auxiliaries AI Agent	Sonnet 5.5	22,000	110,000 (5 tasks)	$0.44
Graphics Agent	Haiku 5.5	20,000	100,000 (5 tasks)	$0.02
UI Agent	Haiku 5.5	20,000	80,000 (4 tasks)	$0.02
Multiplayer Sync Agent	Opus 5.5	28,000	140,000 (5 tasks)	$1.12
Playtest Agent	Sonnet 5.5	20,000	160,000 (8 tasks)	$0.64
Collision and Terrain Agent	Sonnet 5.5	22,000	110,000 (5 tasks)	$0.44
Session and Join Agent	Sonnet 5.5	20,000	80,000 (4 tasks)	$0.32
Audio Agent	Haiku 5.5	15,000	60,000 (4 tasks)	$0.01
Deployment Agent	Sonnet 5.5	20,000	80,000 (4 tasks)	$0.32
Effects Agent	Haiku 5.5	20,000	80,000 (4 tasks)	$0.02
Balance Agent	Sonnet 5.5	20,000	80,000 (4 tasks)	$0.32
Total			**1,640,000**	**$6.12**

A 15% reserve of about 246,000 tokens covers rework after playtesting, bringing the plan to roughly 1.9 million tokens and about $7.04.

Cost uses Anthropic's list API prices per million input/output tokens as of October 2026 (Opus 5.5 $4/$20, Sonnet 5.5 $2/$10, Haiku 5.5 $0.10/$0.50) and assumes 3 input tokens for every output token with no caching or batch discount.

How the work is sequenced

1. **Wave 1:** Lead, Game Logic, World Map, Game Control, and Collision and Terrain build the rules and the world.
2. **Wave 2:** Wrecker AI, Auxiliaries AI, Multiplayer Sync, Session and Join, and Deployment build on Wave 1's output.
3. **Wave 3:** Graphics, UI, Audio and Effects add the look and sound on top of working rules.
4. **Wave 4:** Playtest and Balance run full 5-round matches, tune the numbers, and pay for fixes from the reserve.

API constraints

- **Each call sees only its own brief.** An agent's context holds its spec, its own code, and the shared game-state sheet, capped near 30,000 tokens. The game-state sheet is a one-page list of what every tank reports (position, team color, energy bars, Cells carried, terrain underfoot). Game Logic owns it, and the Wrecker AI Agent can only write "hunt the Cell leader" because that sheet exposes Cells carried.
- **There is no memory between calls.** Every session ends with a handoff note of 500 words or less to the Lead Agent, so work is not repeated or lost.
- **Rate limits cap requests and tokens per minute.** Sixteen agents running at once would hit them, so we run no more than five together, which is why we split the work into waves.
- **The live game cannot call the API.** With up to 28 Wreckers each making 10 decisions per second, the game would need about 280 calls per second, and each call takes seconds. Wrecker and Auxiliary behavior is therefore fixed rule-based code on the game server, and the agents are used only to write and test it.
- **Server target for 20 players.** The Multiplayer Sync Agent aims for about 20 state updates per second to every player, which a 20-player cap keeps affordable.

Tools

Frontend: Angular, Geckos.io (client)

Backend: TypeScript, Node.js/ExpressJS, Geckos.io (server)

Game Framework: Phaser